

1. 請框出答案. 2. 不可使用手機、計算器，禁止作弊!

1. Let $\vec{u} = [2, 3, 1]$, $\vec{v} = [1, 2, x]$. Find all the possible x such that $\vec{u}$ and $\vec{v}$ is perpendicular.

$$x = \underline{-8}$$

Solution :

Find $\vec{v}$ such that $0 = \vec{u} \cdot \vec{v} = 2 \times 1 + 3 \times 2 + 1 \times x$. Therefore, $x = -8$.

2. Prove or disprove that for a vector $\vec{v}$ in $\mathbb{R}^n$, the magnitude of r times $\vec{v}$ is r times the magnitude of $\vec{v}$.

Solution :

The statement is False! Check Section 1-2 problem 40(i)

3. Given two matrices

$$A = \begin{bmatrix} -2 & 1 & 5 \\ 0 & 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 1 & -3 \\ 1 & -2 & 0 \end{bmatrix}$$

Compute the indicated quantity, if it is defined

(a) $(3A)(2B) = \underline{\hspace{2cm}}$

(b) $BA = \underline{\hspace{2cm}}$

Solution :

兩題都不用算！因為沒有定義！